Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

BASICS OF CELL AND MOLECULAR BIOLOGY [BME 3125]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Two proteolytic enzymes, Enzyme A and Enzyme B, have Michaelis constant (K_m), values of 0.5 mM and 2.0 mM, respectively. Analyse what these Km values reveal about each enzyme's substrate affinity and catalytic effectiveness at different substrate concentrations. (3)

1B) How do microfilaments contribute to cell shape, motility, and intracellular transport? Explain their structure and specific roles within the cytoskeleton. (3)

1C) Compare the roles of leading and lagging strands in DNA replication and analyse how they differ in replication mechanisms. (4)

2A) Analyse the potential impact of a mutation in the RB gene on the regulation of the cell cycle. How would this mutation affect cell division and contribute to the development of cancer? (2)

2B) Explain the process of base excision repair in DNA. How does this mechanism correct damaged or mutated bases, and why is it essential for maintaining genomic stability? (3)

2C) Describe the key components involved in DNA transcription, including RNA polymerase, promoters, transcription factors, and enhancers. How do these elements work together to regulate gene expression? (5)

3A) Given the following mRNA sequence:

5' - AUG GCU UUA GCU UGA - 3' (3)

Translate the provided mRNA sequence into the corresponding peptide sequence.

3B) Examine how antibiotics such as tetracycline and erythromycin inhibit bacterial translation. Compare their mechanisms of action and discuss the potential consequences of these inhibitions on bacterial cell function and survival. (3)

3C) Describe the process of translation in protein synthesis. How do ribosomes, tRNA, and mRNA work together to form a polypeptide chain? (4)

4A) Given the pre-mRNA sequence (5' - Exon 1 - Intron 1 - Exon 2 - Intron 2 - Exon 3 - (3)

Intron 3 - Exon 4 - 3'), compare the potential protein isoforms produced through alternative splicing. How might different inclusions or exclusions of exons affect the function of the resulting protein?

4B) What are the key steps involved in isolating DNA from a cell? Describe the purpose of each step and explain how different techniques help to ensure purity and yield of the extracted DNA (3)

4C) How can Western blotting be used to determine the relative levels of protein expression in different samples? Explain how variations in band intensity can indicate differences in protein abundance across samples. (4)

5A) Given the following RT-PCR Ct values for a specific gene:
Gene in Patient 1: C_t value of 18
Gene in Patient 2: C_t value of 35
Compare the Ct values for the gene in these patients. What do the differences in Ct values suggest about the expression level of the gene in each patient? (3)

5B) How does recombinant DNA technology enable the production of proteins like insulin? Describe the key steps involved in isolating the gene of interest, cloning it into a plasmid, transforming bacteria, and expressing the protein. (4)

5C) What are the key characteristics of cancer cells, and how do features like uncontrolled growth and evasion of cell death contribute to tumorigenesis? (3)

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